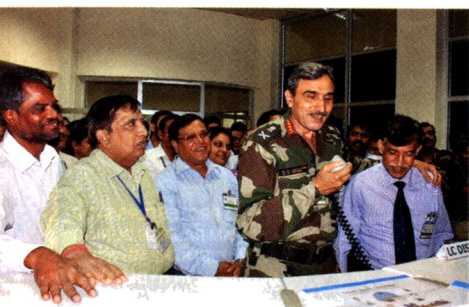
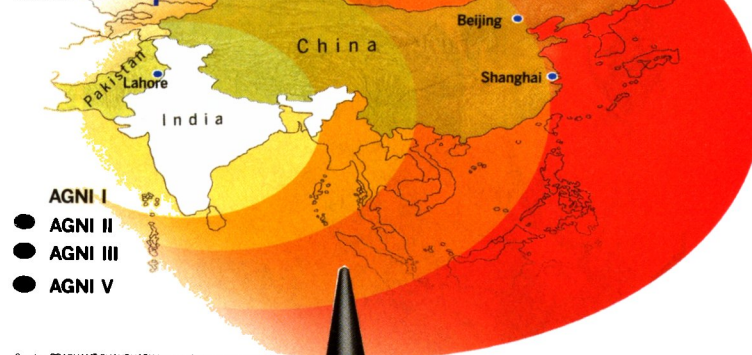




India's Nuclear Strike Envelope



AGNI V

Range: 5,000 km
Weight: Not Available
Height: 17 metres
Engine diameter: 2 metres
Warhead weight: 1,500 kg
Target: China
Tests Done: None
Status: First test in 2011



"India now has a high level of maturity in its ballistic missile arsenal and can deter its adversaries."

V.K. SARASWAT,
DRDO Chief



India's missiles if it is fired from Assam. What India does need is a longer range missile of around 5,000 km to target every corner of China if it wants to apart from firing it from any point in India. For that it is developing Agni V, a three-stage missile system, which is to be launched early next year. The configuration being planned so far is to go with the two stages that Agni III already has as these are flight validated. A third stage would be developed and added that would give the missile the necessary boost to travel an additional 2,000 km. That, says Saraswat, would "give India the full complement of both short and intermediate range ballistic missiles".

Meanwhile, scientists have already identified the glitches that have marred the two flights of the Agni II variant last year. Chander says they have determined that "in both instances these

were caused by quality-related causes". In one flight, the power short-circuited because of poor quality wiring. And in the other, there were problems with the control motors because of shoddy manufacturing. The missile team has now taken steps to streamline production systems and impose strict quality control measures on the 150 industries—both public and private—they rely on to build the Agni systems. In the meantime, scientists are also working on a variant of Agni II to make it road mobile. It is already rail mobile. This is necessary because it would allow the defence forces to transport the missile to any part of the country and, if required, with stealth.

Meanwhile, DRDO is also readying India's ballistic missile defence (BMD) systems. Saraswat, who was earlier its programme director, said they had already validated much of its systems

in test flights, demonstrating capability to intercept incoming ballistic missiles both in the endo- and exo-atmosphere, including high-altitude interceptions. Though the BMD system is expensive, it would give India's key cities the much-needed shield against nuclear missile attacks from Pakistan.

If India has to develop full range of missile systems that China has, then it would also have to build an Inter-Continent Ballistic Missile (ICBM). Its space vehicles, designed and assembled by the Indian Space Research Organisation, already have the rocket power to launch payloads to over 10,000 km. But the government has been wary of converting it into an ICBM or getting DRDO to develop an ICBM on its own. If India ever needed to do so, the DRDO says it could deliver such a missile in a span of five years. That indeed is a reassuring thought for India. ■